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Assessment: MAT9004_Assignment 1

MAT9004 Assignment 1

Due: Friday, 16 August 2024, 23:55 PM (End of Week 4)

Q1. [10 points] Product Pricing: A manufacturer of nails notes that most customers buy a single pack of nails at a time, but will buy a bigger pack if it is cheaper per nail. A marketing company was engaged to conduct a survey and found the following prices that customers were willing to pay for packs of nails of different sizes:

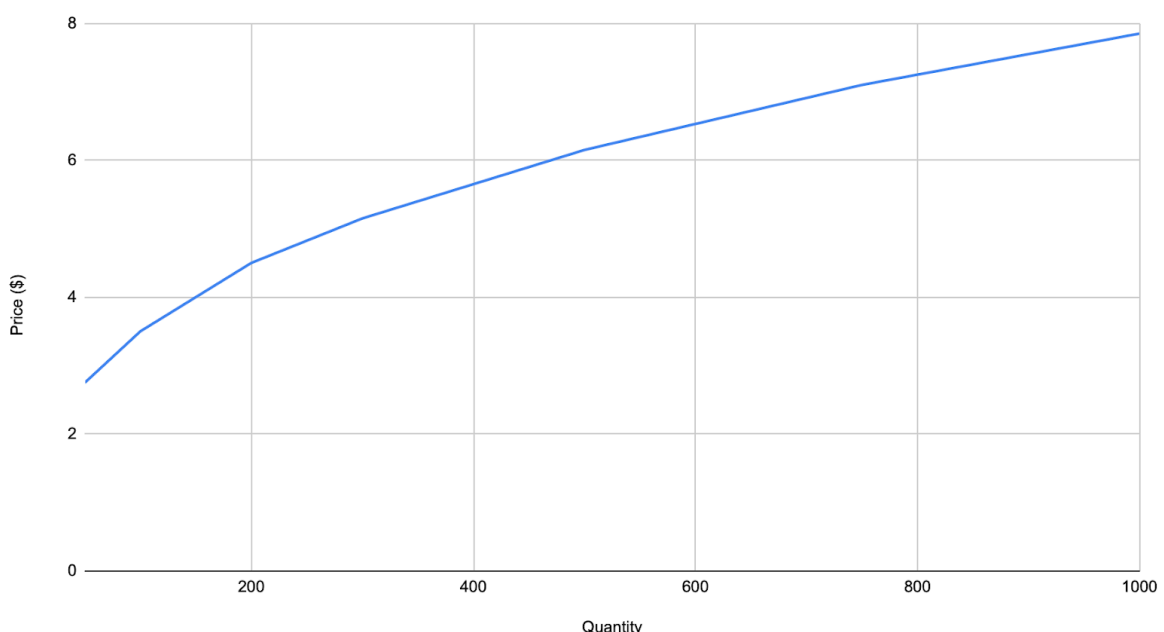
n.nails: 50 100 200 300 500 750 1000

price(\$): 2.75 3.50 4.45 5.15 6.15 7.10 7.85

(a) [3 points] Of the three basic functions for data analyses discussed in the lectures (power law, ex-pponential and logarithmic), which might apply for this data? Choose which you think is best and provide reasons. Include at least one plot with your answer.

Ans.

Price (\$) vs. Quantity



The data analyses that apply for this data would be Logarithmic functions as :

1. The Function is a concave function as for any two points on the line the plot goes over the two joining points.
2. The plot also is visually increasing in one direction.

2 b) As we have considered the plot to be a logarithmic function fitting. we can use the lin-log transformation.

$\therefore$ the function can be written as

$$p(q) = a \cdot \ln(q) + b \quad \therefore \text{ in } y = mx + c$$

$$m = a$$

$$c = b$$

Using the given values of the function =

$$2.75 = a \cdot \ln(50) + b$$

$$2.75 - a \cdot \ln(50) = b \quad \text{--- (i)}$$

using another value:-

$$3.5 = a \cdot \ln(100) + b$$

$$3.5 - a \cdot \ln(100) = b \quad \text{--- (ii)}$$

Using (i) & (ii)

$$2.75 - a \cdot \ln(50) = 3.5 - a \cdot \ln(100)$$

$$a(\ln(100) - \ln(50)) = 3.5 - 2.75$$

$$a = \frac{(3.5 - 2.75)}{\ln(100) - \ln(50)} = \frac{0.75}{0.69} = 1.08$$

$$\therefore a \approx 1.08$$

$$\text{for } b \Rightarrow 3.5 - 1.08 \cdot \ln(100) = b$$

$$b \approx -1.47$$

$\therefore$ the coefficients are $a \approx 1.08$
 $b \approx -1.47$

$$\therefore p(q) = 1.08 \ln(q) - 1.47$$

c) $\therefore$ to calculate the cost: (Here $f(q) = \text{prof}(q)$
both functions give the profit)

$$\text{cost} = \cancel{0.3 \times q} +$$

$$\text{cost (in dollars)} = (0.003 \times q) + 0.08$$

~~pro~~ $\therefore$ the profit is price - cost

$$p(q) = 1.08 \ln(q) - 1.47$$

$$c(q) = (0.003 \times q) + 0.08$$

$$\therefore \text{profit} = (1.08 \ln(q) - 1.47) - ((0.003q) + 0.08)$$

$$\text{prof}(q) = (1.08 \ln(q) - 0.003q) + (-1.47 - 0.08)$$

$$= 1.08 \ln(q) - 0.003q - 1.55$$

deriving this function

$$\text{prof}'(q) = \frac{d}{dq} (1.08 \ln(q) - 0.003q - 1.55)$$

$$\text{prof}'(q) = \frac{1.08}{q} - 0.003$$

$\Rightarrow$ to find the maximum profit we equate $\text{prof}'(q)$ to 0

$$\therefore \text{prof}'(q) = 0$$

$$\frac{1.08}{q} - 0.003 = 0$$

$$\frac{1.08}{0.003} = q$$

$$360 = q$$

$\therefore$ for a packed size of 360 nails there will be maximum profit.

$$\begin{aligned}\text{prof}(q) &= 1.08 \cdot \ln(360) - 0.003 \cdot (360) - 1.55 \\ &= 0.13\end{aligned}$$

$\therefore$ A profit of \$0.13.

d) If $f(q) = 1.08 \cdot \ln(q) - 0.003 \cdot q - 1.55$

Domain :-

for any logarithmic function $\ln(q)$

$$q > 0 \text{ i.e. +ve}$$

Range:- for the term $1.08 \cdot \ln(q) - 0.003 \cdot q - 1.55$

as q increases the first term increases i.e. $1.08 \cdot \ln(q)$

as the term q increases $-0.003q$ decreases.

$\therefore$ range can vary for $[-\infty, \infty]$

$\therefore$ Domain :- $q > 0$

Range :- $[-\infty, \infty]$

MAT9004 Assignment - 1

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$$2. a) \quad g(x) = \frac{5}{18} x^3 + \frac{1}{6} x^2 + 5 - \frac{1}{\ln(2)} 2^x$$

$$\begin{aligned} g'(x) &= 3 \times \frac{5}{18} x^2 + 2 \times \frac{1}{6} x + 0 - \frac{\ln(2)}{\ln(2)} \times 2^x \\ &= \frac{5}{6} x^2 + \frac{1}{3} x + 2^x \end{aligned}$$

b) A function is said to be invertible if it is both injective and surjective, i.e. if it is a bijective function.

Checking $g'(x)$ at the point $x=1$ to check strictly inc. function

$$\begin{aligned} g'(1) &= \frac{5}{6} \cdot 1^2 + \frac{1}{3} + 2 \\ &= \frac{5}{6} + \frac{1}{3} + 2 \\ &= \frac{7}{6} + 2 = +ve \end{aligned}$$

To check estimated range around $x=1$, let's check for $x=0.8$ &

$$g'(0.8) = \frac{5}{6} (0.8)^2 + \frac{1}{3} (0.8) + 2^{(0.8)}$$

= This value is always +ve

$$g'(1.2) = \frac{5}{6} (1.2)^2 + \frac{1}{3} (1.2) + 2^{(1.2)}$$

$\Rightarrow$ Here also every value is +ve.

$\therefore$ for an estimated range of $[0.8, 1.2]$ the function is invertible

$$c) \quad g(x) = \frac{5}{18} x^3 + \frac{1}{6} x^2 + 5 - \frac{1}{\ln(2)} 2^x$$

$$g'(x) = \frac{5}{6} x^2 + \frac{x}{3} + 2^x$$

$$g''(x) = \frac{5}{3} x + \frac{1}{3} + \ln(2) 2^x$$

now checking the 2nd derivative of g over the ~~value~~ interval $[0, 1]$ to check for convex

$\Rightarrow f$ is convex if and only if $f''(x) \geq 0$ for all $x \in K$

$\therefore$ for $x = 0$ i.e. boundary point

$$g''(x) = \frac{1}{3} + \ln(2)$$

which is always +ve.

for $x = 1$ i.e. boundary point

$$g''(x) = \frac{5}{3} + \frac{1}{3} + \ln(2) \cdot 2$$

which again is always positive.

$\therefore$ the function $g(x)$ is convex over the interval $[0, 1]$

3. Let's assume that the ambulance station is x distance away from the first town.

$\therefore$ If all the towns are connected by a highway the distance from each town can be represented by: ~~$|d_i - x|$~~
 $|x - d_i|$ where d_i is the location of town i .

a) $\therefore$ for the distance to be minimised.

The function can be defined as:

$$f(x) = \frac{1}{n} \sum_{i=1}^n (x - d_i)^2$$

This denotes the average square of the distance between towns and ambulance station.

To find the optimal x in terms of the d_i the function is:

$$\begin{aligned} f(x) &= \frac{1}{n} \sum_{i=1}^n (x^2 - 2xd_i + d_i^2) \\ &= \frac{1}{n} \left(\sum_{i=1}^n x^2 - 2x \sum_{i=1}^n d_i + \sum_{i=1}^n d_i^2 \right) \\ &= \frac{1}{n} \sum_{i=1}^n x^2 - \frac{2x}{n} \sum_{i=1}^n d_i + \frac{1}{n} \sum_{i=1}^n d_i^2 \end{aligned}$$

now using $\frac{d}{dx} f(x) = 0$

$$\frac{2}{n} \sum_{i=1}^n x - \frac{2}{n} \sum_{i=1}^n d_i + 0 = 0$$

2) Since $\frac{2}{n} \sum_{i=1}^n x$ does not have a term with dependent

$$\text{on } i \quad \frac{2}{n} \sum_{i=1}^n x = \frac{2}{n} \times nx = 2x$$

$$2u - \frac{2}{n} \sum_{i=1}^n d_i = 0$$

$$2u = \frac{2}{n} \sum_{i=1}^n d_i$$

$$u = \frac{1}{n} \sum_{i=1}^n d_i$$

b) To calculate the average distance between the station and all cities we can calculate:

$$g(u) = \frac{1}{n} \sum_{i=1}^n (u - d_i)$$

where u is the distance from the first town and d_i is the distance between of the town i

Now to calculate the minimum of the function $g(u)$ we need $g'(u) = 0$

$$g'(u) = \frac{1}{n} \sum_{i=1}^n (1 - 1) = 0$$

$$g(u) = \frac{1}{n} \sum_{i=1}^n u - \sum_{i=1}^n d_i = u - \frac{1}{n} \sum_{i=1}^n d_i$$

$$g'(u) = \frac{1}{n} \cdot n - \frac{1}{n} \sum_{i=1}^n 0$$

$$= 1 - \frac{1}{n} \sum_{i=1}^n 0$$

Here $g'(u)$ cannot be calculated as the term u will not be ~~calc~~ in the derived eqⁿ.

3c) To find the distance between the helipad and the towns :-
 (u_i, v_i) (x, y)

$$\Rightarrow \sqrt{(x - u_i)^2 + (y - v_i)^2}$$

Now, to find the population average of the squared distances :-

i.e. sum of population size times squared distance all divided by the total population :-

$$\Rightarrow \frac{1}{\sum P_i} \sum_{i=1}^n P_i [(x - u_i)^2 + (y - v_i)^2]$$

$$\Rightarrow \frac{1}{\sum P_i} \left[\sum_{i=1}^n P_i (x - u_i)^2 + \sum_{i=1}^n P_i (y - v_i)^2 \right] \quad \text{--- ①}$$

Where P_i is the population of town i

i) Showing it as the sum of independent functions :-

$$\approx \text{i.e.} \Rightarrow f(x) + g(y)$$

using ①

$$f(x) = \sum_{i=1}^n P_i (x - u_i)^2$$

$$g(y) = \sum_{i=1}^n P_i (y - v_i)^2$$

$$\text{Dist}(x, y) = \frac{1}{\sum P_i} [f(x) + g(y)]$$

ii) Derivating both $f(x)$ & $g(y)$ individually :-

$$\Rightarrow f(x) = \sum_{i=1}^n p_i (x - u_i)^2$$

$$\frac{df(x)}{dx} = \sum_{i=1}^n p_i \cdot 2(x - u_i)$$

$$= 2 \sum_{i=1}^n p_i (x - u_i)$$

now minimizing $f(x)$ $\Rightarrow$

$$2 \sum_{i=1}^n p_i (x - u_i) = 0$$

$$\sum_{i=1}^n p_i x = \sum_{i=1}^n p_i u_i$$

$$x = \frac{\sum_{i=1}^n p_i u_i}{\sum_{i=1}^n p_i}$$

Similarly :-

$$y = \frac{\sum_{i=1}^n p_i v_i}{\sum_{i=1}^n p_i}$$